Scientific Notation

Prodigy Anesthesia

Converting Numbers and Decimals into Scientific Notation

Scientific notation combines the decimal system with exponents to shorten the written form of longer numbers. For example, the number 1000 can be written as

$$1 \text{ X } 10^3$$

In this expression, the exponential portion 10^3 is the same as writing 10 X 10 X 10 or 1000. Although in this example, it doesn't seem to be a shortcut, the advantage becomes more apparent with more complex numbers. For example, the number 2,345,000,000,000 could be written as

$$2.345 \text{ X } 10^{12}$$

The exponent portion of the expression (12) is determined by counting the number of spaces that occur to the right of the decimal point. Scientific notation is always written with one number to the left of the decimal point. To help you better understand the process, we'll practice converting a few other large numbers to scientific notation over the next few pages. An interesting note is that the exponent 10^0 is considered to be the same as the number 1. So, the expression 7.0 X 10^0 is the same as 7 X 1 or 7. In short, any number X 10^0 is simply that number.

Converting Numbers Into Scientific Notation

Convert the number 67,546 into scientific notation

Step One: Identify the first number in the expression and write it down.
The first number of 67,546 is 6.

6

Step Two: Scientific notation always appears in the form of one number to the left of the decimal, so we can write the decimal down as well.

6.

Step Three: Next, write the numbers down that follow the six. We won't always write them all down, but that involves a term called significant figures and we'll get to that later.

6.7546

Step Four: The next portion of the expression is the exponent, which is indicated by X 10 to indicate that we are going to multiple this number by a series of tens, so write X 10 down.

6.7546 X 10

Step Five: Remember earlier I indicated that scientific notation is the same as multiplying a decimal by a series of tens? The number of tens we multiply our decimal (6.7546) by is determined by the number of digits to the right of the decimal. The four numbers 7546 appear to the right of the decimal, so we must multiply 6.7546 by four tens (10 X 10 X 10 X 10). A shorter way to write these four tens is by writing the exponent 4 as a superscript to the right of the 10. So, our final result is:

$$6.7546 \text{ X } 10^4$$

Now, let's try converting the number 2,105 into scientific notation.

We'll start by writing down the first number and the decimal.

2.

Next, write down the numbers that follow the 2. Also, write down the X 10 portion of the exponent.

2.105 X 10

Now, we need to figure out the number to write down as the exponent. Do this by counting the number of digits that follow the 2. There are three numbers (105), so write the number 3 as the exponent.

$$2.105 \text{ X } 10^3$$

That's it. We're finished. Well, almost. We still have to address numbers that are smaller than one. For example:

Convert the number 0.0098677 into scientific notation.

Step One: We need to identify the first number to the left again. Only this time, ignore the zeroes that appear on the left-hand side of the number. We'll get to them later. We find that if we ignore the zeroes, the first non-zero number we find starting from the left is 9. So, let's write it down.

9

Step Two: As we said earlier, all scientific notation appears in the form of a single number, followed by a decimal point, followed by some more numbers (usually), followed by the exponent expression which looks like X 10^n where n is the exponent. So, it looks like we need to write a decimal point, and then write the rest of the numbers that follow the 9.

9.8677

Step Three: Now, we need to write the easy part—the X 10. We know that every scientific notation is going to have this, so it doesn't really require any figuring.

9.8677 X 10

Step Four: Only one part remains—the exponent. Earlier, we counted the number of digits to the right of the nine. This time it's a little different. We will count the number of digits to the right of the original decimal point including the zeroes and our 9 that we used to start our scientific notation. Confused? Don't worry, I'll explain.

Earlier, I said we would get to the zeroes. Now, we need to add them back to the number (at least in our imagination) so we can determine the exponent. To make it a little easier, let's put the zeroes back on our number. One point I should make is that we are only going to really focus on the zeroes that appeared between the decimal point and the 9. Looking at our original number of 0.0098677, it appears that there are two zeroes between the decimal point and the 9. I've underlined them for you. I'll explain why I underlined the 9 also in a just a moment. First, I probably need to explain why we don't worry about the zero to the left of the decimal point.

Why do we ignore the zero to the left of the decimal? Because no matter how many zeroes appear to the left of the decimal, as long as there are no digits bigger than zero, it has no effect on the number. For example, 0.00998677 is the same as 000.0098677 and .0098677. Whether there are 5 zeroes to the left of the decimal or no zeroes to the left of the decimal, it doesn't make the number larger or smaller.

So, back to our question. Our goal in finding out the exponent is to determine how many spaces we would need to move our original decimal point to put it where we find it in the scientific notation. For example, we went from 0.0098677 to 9.8677 X 10. We moved our decimal point from the left of the 9

0.0098677

to the right of the nine. How many spaces did we move it? Let's see:

It appears that we moved the decimal point three spaces to the right to create our scientific notation. That means that our exponent is 3.

Important Note: Because our number is less than 1, we need to use a negative exponent!

So, our finished expression should be written:

$$0.000634$$

$$9.8677 \text{ X } 10^{-3}$$

Let's do another one to make sure you have it. This time, we'll combine a few steps.

Convert the number 0.000634 into scientific notation.

Step One: Find the first non-zero number to the right of the decimal. Since we know we always follow it with a decimal point, write it down, too.

$$6.$$

Step Two: Now, write down the rest of the numbers that appear after the six. Since we know that we always add the X 10 to scientific notation, we can write it down as well.

$$6.34 \text{ X } 10$$

Step Three: Now all that remains is to count how many spaces we moved our decimal point from the original number to where it appears in our scientific notation and that number will be our exponent. Our original number was 0.000634 and our scientific notation looks like 6.34 X 10 so far. To get the decimal point from where it started to the right of the six we just count the spaces in between:

It appears that we moved the decimal four spaces, so our exponent is 4.

Write it down, and you are finished:

$$6.34 \text{ X } 10^{-4}$$

Converting Scientific Notation Into a Number or Decimal

Now, that you know how to change a number or decimal into scientific notation, converting scientific notation back into standard form is fairly easy. All we have to do is shift the decimal point to the left or right according to our exponent. To make it even easier, we'll break it into a few steps.

For example:

Convert $2.345 \text{ X } 10^3$ into decimal (number) form.

Step One: Write the number down as it appears above. This is our starting point

$$2.345 \text{ X } 10^3$$

Look at the exponent above the 10. Is it positive or negative? This one is positive. That means we will shift the decimal to the right. If our exponent was negative, we would move our decimal to the left.

Step Two: What number is the exponent? This one is 3. That means we will shift the exponent three places to the right. Now that we know how far to move the decimal point and in what direction, we can remove the $\text{X } 10^3$ portion. That's only relevant to scientific notation and we won't need it.

$$2.345$$

Step Three: Move the decimal point. Remember, we said we're moving it three places to the right, so I'll show you each step.

Start:	2.345
Decimal moved one place	23.45
Decimal moved two places	234.5
Decimal moved three places	2345.0

That's it. We're finished. We converted $2.345 \text{ X } 10^3$ into standard form and our answer is 2345. Let's try another.

Convert 1.2908×10^{-4} into standard decimal form.

Step One: Write the number down as it appears above so we have our starting point.

$$1.2908 \times 10^{-4}$$

Step Two: Now look at the exponent. This one is negative four. That means we will shift our decimal point four places. Because the exponent is negative, we will shift it to the left. Let's go ahead and remove the $\times 10^{-4}$ portion since we don't need it anymore.

$$1.2908$$

Step Three: Shift the decimal point four places to the left. We will fill in zeroes where we have blanks. We'll do it in steps again.

Start:	1.2908
Decimal moved one place	0.12908
Moved two places	0.012908
Moved three places	0.0012908
Moved four places	0.00012908

We're finished. We converted 1.2908×10^{-4} into standard form by moving the decimal point four places to the left to get the final answer 0.00012908.

Scientific Notation Practice Problems Part A

Convert the following numbers into scientific notation:

1. 101,375
2. 47
3. 0.983
4. 154
5. 1,213
6. 0.0015
7. 1.5
8. 860
9. 860
10. 4,201
11. 1,000,015
12. 0.2
13. 5.97
14. 135.98
15. 503
16. 1,956
17. 6,015.3
18. 27,000
19. 32
20. 18,123
21. 41.3
22. 0.00005
23. 0.0213
24. 0.0321658
25. 1,100
26. 110,101
27. 323
28. 4,201
29. 9,137,000
30. 816,245
31. 27,060
32. 3200
33. 32,000
34. 57,860.9
35. 40.04
36. 0.00006
37. 0.315
38. 11.12
39. 293
40. 7
41. 700
42. 154,003
43. 98
44. 67,250.1
45. 140
46. 1,000,000
47. 10.1836
48. 0.014
49. 19,190
50. 5,632,114.3

Scientific Notation Practice Problems Part B

1. 1.4×10^{2}
2. 6.8×10^{-3}
3. 4.32105×10^{0}
4. 2.2×10^{-6}
5. 8.8403×10^{4}
6. 3.0×10^{3}
7. 9.99909×10^{7}
8. 9.9909×10^{-7}
9. 7.0004×10^{1}
10. 3.0×10^{-3}
11. 1.0101×10^{3}
12. 2.83×10^{2}
13. 5.0×10^{5}
14. 1.821643×10^{3}
15. 3.34×10^{-2}
16. 4.0004×10^{4}
17. 1.5×10^{-6}
18. 6.123×10^{-1}
19. 1.1×10^{11}
20. 5.0312×10^{-2}
21. 2.914×10^{8}
22. 8.49168321×10^{3}
23. 3.001013×10^{-3}
24. 2.0×10^{0}
25. 1.0×10^{3}
26. 5.516×10^{-4}
27. 7.777×10^{3}
28. 8.1818×10^{-4}
29. 3.230456×10^{1}
30. 1.83×10^{-1}
31. 4.6×10^{3}
32. 6.813×10^{2}
33. 2.222×10^{2}
34. 3.1×10^{-3}
35. 8.23×10^{8}
36. 5.5×10^{5}
37. 9.08×10^{-1}
38. 7.1×10^{12}
39. 2.41×10^{6}
40. 6.0×10^{6}
41. 5.12×10^{-6}
42. 4.01×10^{4}
43. 1.93×10^{3}
44. 8.88×10^{-8}
45. 3.1415×10^{0}
46. 7.98143×10^{2}
47. 2.2202×10^{2}
48. 1.1×10^{-1}
49. 6.8×10^{0}
50. 9.681358×10^{4}

Scientific Notation Answers Part A

1. 1.01375×10^{5}
2. 4.7×10^{1}
3. 9.83×10^{-1}
4. 1.54×10^{2}
5. 1.213×10^{3}
6. 1.5×10^{-3}
7. 3.4×10^{-2}
8. 1.5×10^{0}
9. 8.6×10^{2}
10. 4.201×10^{3}
11. 1.000015×10^{6}
12. 2.0×10^{-1}
13. 5.97×10^{0}
14. 1.3598×10^{2}
15. 5.03×10^{2}
16. 1.956×10^{3}
17. 6.0153×10^{3}
18. 2.7×10^{4}
19. 3.2×10^{1}
20. 1.8123×10^{4}
21. 4.13×10^{1}
22. 5.0×10^{-5}
23. 2.13×10^{-2}
24. 3.21658×10^{-2}
25. 1.1×10^{3}
26. 1.10101×10^{5}
27. 3.23×10^{2}
28. 4.201×10^{3}
29. 9.137×10^{6}
30. 8.16245×10^{5}
31. 2.706×10^{4}
32. 3.2×10^{3}
33. 3.2001×10^{4}
34. 5.78609×10^{4}
35. 4.004×10^{1}
36. 6.0×10^{-5}
37. 3.15×10^{-1}
38. 1.112×10^{1}
39. 2.93×10^{2}
40. 7.0×10^{0}
41. 7.0×10^{2}
42. 1.54003×10^{5}
43. 9.8×10^{1}
44. 6.72501×10^{4}
45. 1.4×10^{2}
46. 1.0×10^{6}
47. 1.01836×10^{1}
48. 1.4×10^{-2}
49. 1.919×10^{4}
50. 5.6321143×10^{6}

Scientific Notation Answers Part B

1. 140
2. 0.0068
3. 4.32105
4. 0.0000022
5. 88,403
6. 3,000
7. 99,990,900
8. 0.00000099909
9. 70.004
10. 0.003
11. 1,010.1
12. 283
13. 500,000
14. 1,821.643
15. 0.0334
16. 40,004
17. 0.0000015
18. 0.6123
19. 110,000,000,000
20. 0.050312
21. 291,400,000
22. 8,491.68321
23. 0.003001013
24. 2
25. 1000
26. 0.0005516
27. 7,777
28. 0.00081818
29. 32.30456
30. 0.183
31. 4,600
32. 681.3
33. 222.2
34. 0.0031
35. 823,000,000
36. 550,000
37. 0.908
38. 7,100,000,000,000
39. 2,410,000
40. 6,000,000
41. 0.00000512
42. 40,100
43. 1,930
44. 0.0000000888
45. 3.1415
46. 798.143
47. 222.02
48. 0.11
49. 6.8
50. 96,813.58